Hydrofoil selection

7 open-source sections (UIUC), polars from NeuralFoil (open-source XFOIL surrogate); each section built into rotors and scored as a whole turbine with the GL60



NACA 4412
t/c 12 % conf. 0.98
SELECTED



NACA 4415
t/c 15 % conf. 0.98
searched



SG6043
t/c 10 % conf. 0.92
searched



FX 63-137
t/c 14 % conf. 0.96
searched



NACA 4418
t/c 18 % conf. 0.97
root section

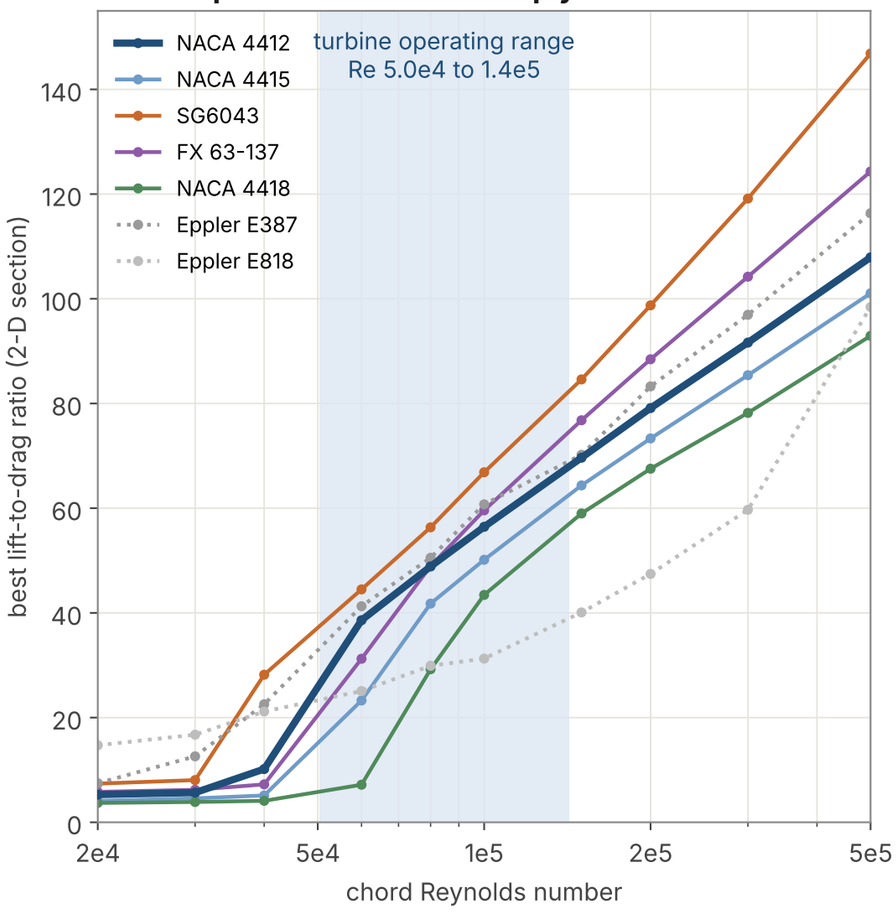


Eppler E387
t/c 9 % conf. 0.84
excluded

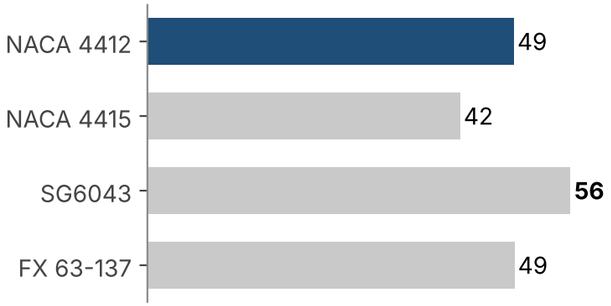


Eppler E818
t/c 12 % conf. 0.64
excluded

Section performance falls steeply below Re 1e5

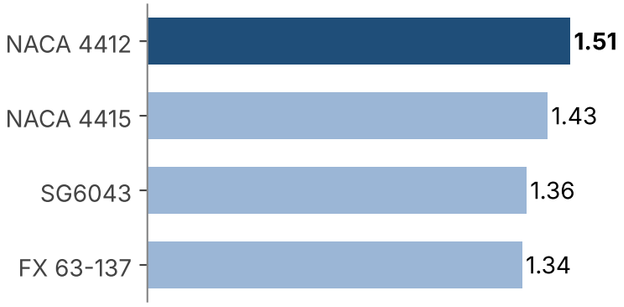


2-D ranking: best L/D at Re 8e4



SG6043 is best in 2-D ...

System ranking: mean net 12 V power 0.5-1.0 m/s, best rotor per section (W)



... but NACA 4412 makes the most electricity

1 Trust gate

surrogate confidence >= 0.90
E387 (0.84), E818 (0.64) excluded

2 2-D performance

L/D over Re 5.0e4 to 1.4e5
SG6043 best: 56 at Re 8e4

3 Decider: system output

508 rotors built from the sections, scored on net power with the GL60

4 Printability

t/c >= 12 % on the blade;
18 % NACA 4418 at the root

Why NACA 4412 won

High-lift sections need narrower chords: lower Re and a sharper Cp peak the generator load cannot stay on.

Design iterations: net power of each whole-turbine design

Dates from the project history. All designs re-evaluated with today's coupled model (BEM + GL60 + buck-boost MPPT) so the curves are comparable. Model predictions; nothing measured yet.

29 Oct
1 Blade for 1-2 m/s

Target
1-2 m/s, 8-in blade
Explored
BEM design; review rev 1 → rev 2
Chose
NACA 44xx, TSR 4, +6° pitch
Result
rotor Cp 0.37 at 1.5 m/s

31 Oct
2 Generator + housing

Target
1-2 m/s, waterproof unit
Explored
8 generators, 4 layouts (A-D)
Chose
GL60 KV25, sealed nacelle
Result
2.5x the 76 mm depth target

31 Oct
3 76 mm redesign (v3.1)

Target
1-2 m/s, ≤ 76 mm deep
Explored
6 rotor variants; HW7 sheet
Chose
motor inside 48 mm drum hub
Result
9.7 W at 1.5 m/s; 0.51 W at 0.7

2 Nov
4 Low-flow optimisation (v4)

Target
0.5-1.0 m/s, max net energy
Explored
508 rotors: 2-5 blades, 4 foils, TSR 1.5-6
Chose
TSR 2.0 ×1.3 → after review TSR 1.5, −3°
Result
0.96 W at 0.7; starts 0.6 m/s

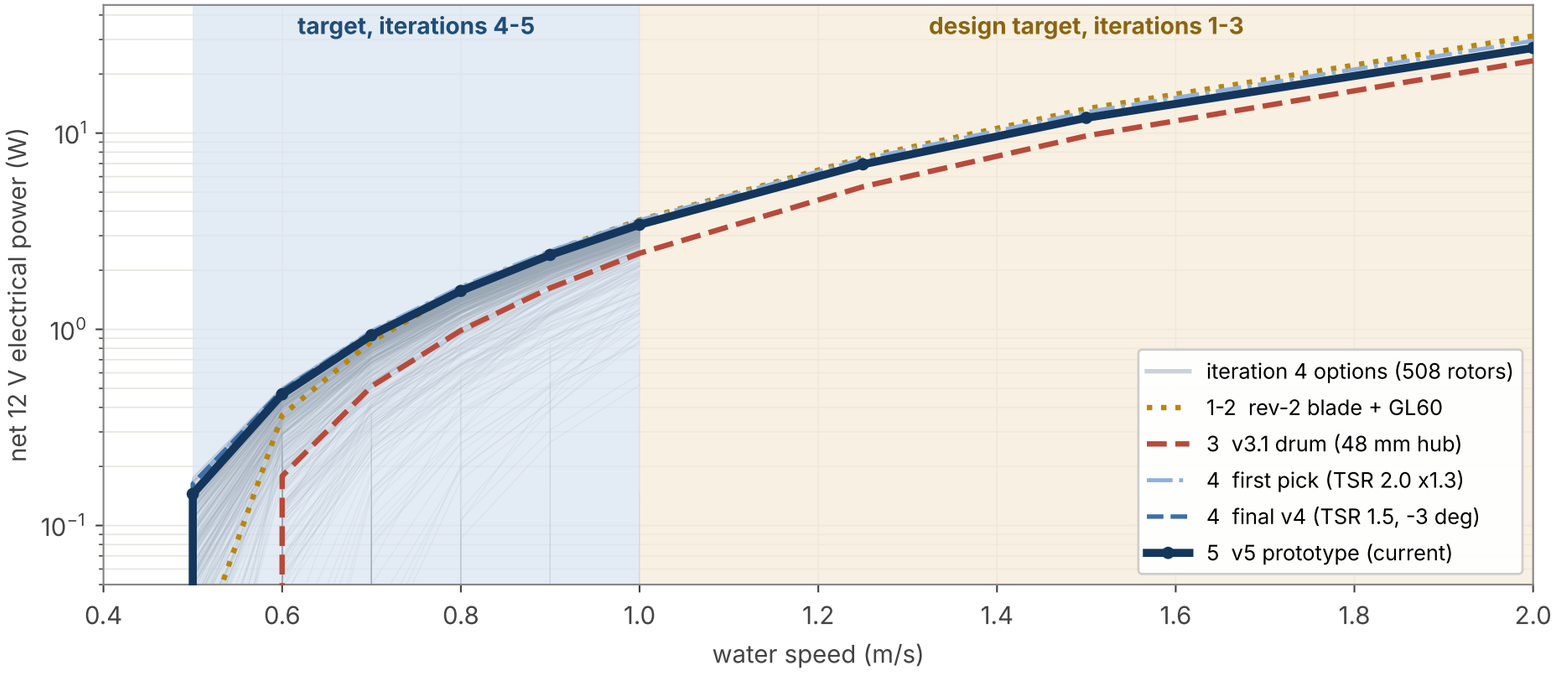
3 Nov
5 Simplified prototype (v5)

Target
printable, fewest parts
Explored
one-piece rotor, housing, mount
Chose
one 12 mm seal + one 6001
Result
0.93 W at 0.7; 91 mm deep

Next
Test and recalibrate

- Bench the GL60: no-load loss, cogging, volts per rpm
- Print the rotor; flume or tow test at 0.5-1.0 m/s
- Refit the model to the data
- If motor losses are high: revisit rotor diameter

PIVOT: flow target lowered to 0.5-1.0 m/s



Net power (W) and self-start speed (m/s)

iteration	0.5	0.7	1.0	1.5 m/s	starts
1-2 rev-2 blade + GL60	0.02	0.87	3.61	13.3	0.7
3 v3.1 drum	0.00	0.51	2.44	9.68	1.0
4 first pick	0.16	0.98	3.60	12.8	0.7
4 final v4	0.16	0.96	3.45	12.0	0.6
5 v5 (current)	**0.14**	**0.93**	**3.41**	**12.0**	**0.6**

Lesson per pivot: the 1-2 m/s designs lose most of their output below 0.8 m/s and start late. Optimising the whole chain for 0.5-1.0 m/s gained 28 % at 0.6 m/s over the rev-2 blade and lowered self-start to 0.6 m/s, at a cost of 13 % at 2 m/s.

Generator: CubeMars GL60 KV25 curve vs how the turbine uses it

Manufacturer curve at 24 VDC (motoring), digitized; the turbine runs it as a generator in the shaded corner

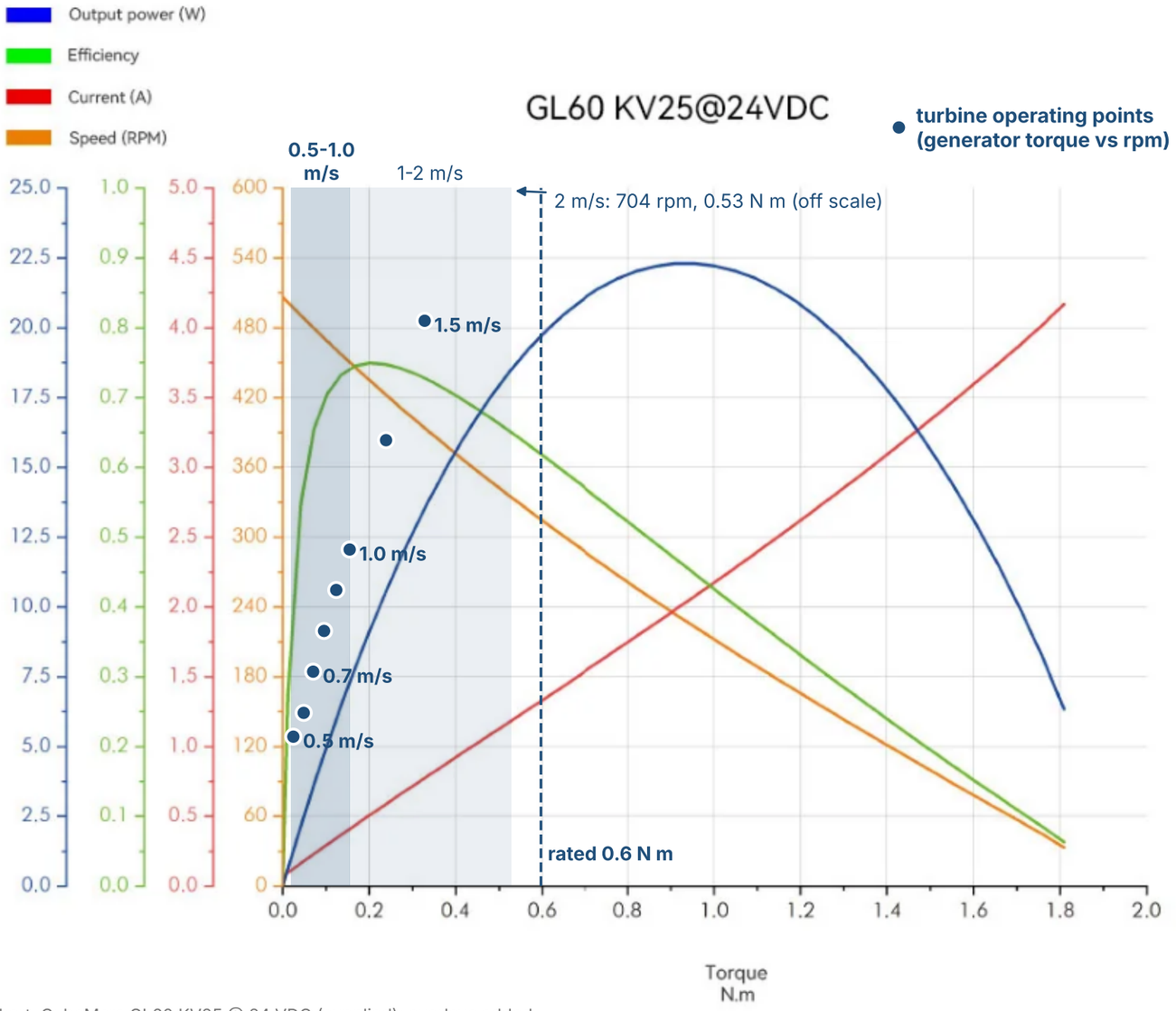


Chart: CubeMars GL60 KV25 @ 24 VDC (supplied); overlays added

Constants check: listed vs chart vs model

quantity	listed	from the chart	model	verdict
speed constant KV	25 rpm/V	516 rpm at 24 V = 21.5	21.5	**match**
torque per amp Kt	0.45 N m/A	0.44-0.48	0.444	**match**
resistance	5.5 ohm	~7.3 ohm effective	5.5	**small effect**
no-load loss	not listed	<= 2.2 W at 516 rpm	0.32 W	**KEY RISK**
turbine load	0.6 N m, 1.35 A	-	0.03-0.16 N m	**4-26 % rated**

KV and Kt confirm the constants behind every prediction. The open question is the chart's 0.09 A at near-zero torque: it may include the controller's own draw, so it is an upper bound on the motor's drag (model assumes 0.32 W at 516 rpm).

Next step: bench-measure the GL60's no-load drag at 100-700 rpm (gate G1).

Effect on the current design's net output

